COMP 3500SEF · Final Report

Warehouse Logistics Management System
Group 42: Tang Yi Jie, Yin Pei Xu, Tang Chao, Yang Ting Zhang, Wu Peng Bo, Chung Ngon Man, Li Mu Qiu.

1. Scope

The goals are traceable order decisions — a ledger entry records each stock movement, and customer-visible events also write notifications — and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. GPS, scanning, telephony, payment and carrier integration are outside scope. Deployment is local-only for one warehouse and four roles. Stock checks, one confirmation transaction (§5) and role gates (per-role permissions, FR-07) guard integrity.

2. Development process

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Coursework kickoff	Scope, responsibilities and contracts — agreed endpoint and data shapes — recorded (increment register)
17 Oct 2026 review	Order and stock functions ready for review	Defects recorded in the increment register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Review found a confirmation endpoint that accepted repeat submits and a missing ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Foundation application integrated	All four role workflows verified in the 11 Nov regression (browser checks)
Driver and privacy, 2 Nov 2026	Driver workflow and data minimisation outstanding	Driver assignment and data-minimisation checks pass (tests)
Documentation hardening, 10–11 Nov 2026	Review found diagrams and traceability diverging from the implementation	Diagrams and traceability aligned (traceability)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date CSV export in three in-app interface interactions (walkthrough record)
CI health tests, 11–12 Nov 2026	The CI health record (per-run outcomes) had no regression test	Two regression tests added; suite reached 38 (unit.log)

3. Requirements

Functional requirements and verification

FR	The system shall	Verification
FR-01	create orders with product lines and totals	Test — creation and stock
FR-02	enforce lifecycle and stock reservation or release	Test — lifecycle, stock and cancellation (cancel tests)
FR-03	number orders and shipments and show owner tracking	Test — identifiers and tracking
FR-04	show inventory, replenishment, ledger and low-stock alerts	Test — inventory, replenishment and alerts
FR-05	record notifications for confirm, ship, pickup, delivery and cancel events and allow reading	Test — notifications and reads
FR-06	provide admin aggregates and CSV export	Test — aggregates and export
FR-07	authenticate and enforce role and object access	Test — authentication and boundaries
FR-08	assign shipped orders and simulate delivery	Test — assignment and delivery

- NFR-01 performance — API p95 ≤2 s, first contentful paint ≤3 s (measurement, §6). - NFR-02 CI health — green build, test and browser runs per CI run (observations, 40 runs). - NFR-03 least privilege and data minimisation — role gates on mutating endpoints; staff responses omit customer identity and delivery address (privacy tests). - NFR-04 interaction budget — at most three in-app interface interactions per core operation; login, navigation, waits, assertions and setup are excluded (walkthrough, §8). - NFR-05 process quality — increments land through green CI runs (observations).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	scheduling plan	increment register
RK-02 contract rework	Lifecycle transitions asserted as constants	transition tests
RK-03 test-week capacity	Traceability	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	Seed data regression-tested	seed regression test
RK-06 coursework assignments and test-week squeeze	Weekly re-planning	risk plan
RK-07 stack unfamiliarity	Setup guide and CI lower ramp-up cost	setup guide
RK-08 report and presentation-video deadlines	Weekly re-planning with video milestones	risk plan

4. System models

The four layers — role pages, HTTP routes and authentication, business handlers, Prisma/SQLite persistence — separate interface, application and data concerns. Figures 1–5 answer, in order: role functions; transitions and stock effects; data relationships; the confirmation transaction; layer connections.

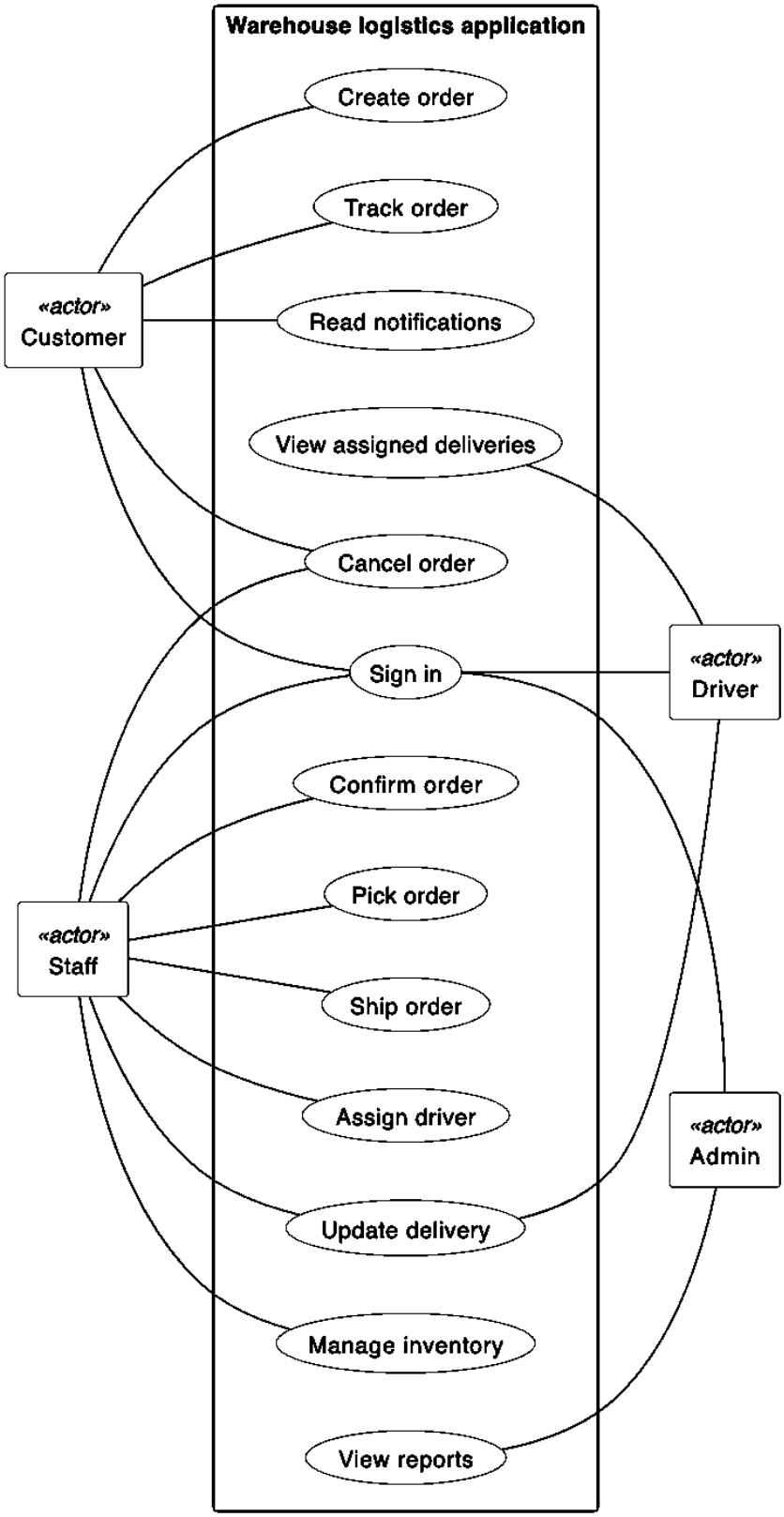


Figure 1. Role functions. Source.

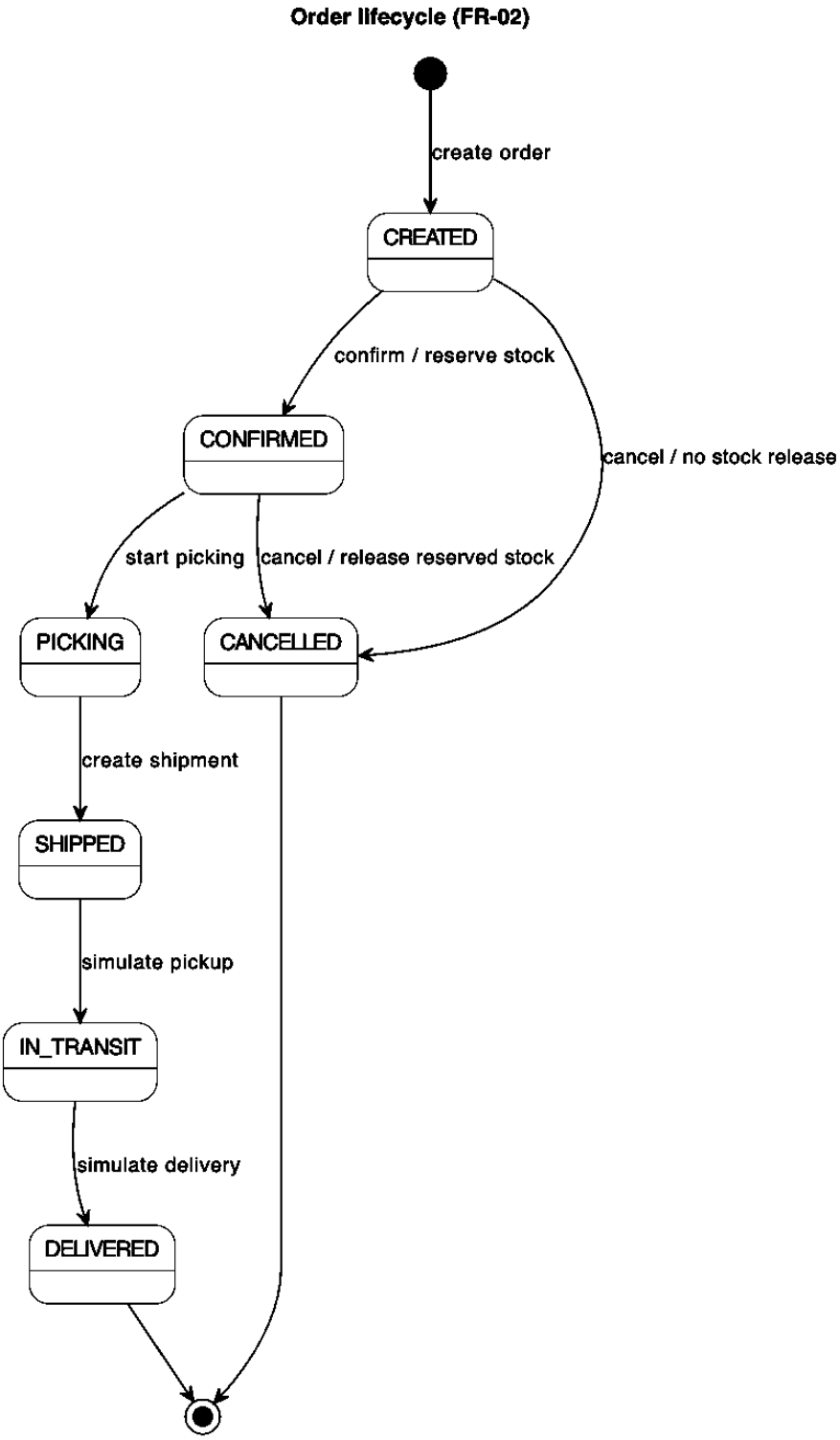


Figure 2. Order transitions and stock effects. Source.

Each item references one product; each order needs one item; orders have 0..1 shipment; products have 0..* ledger entries; notifications relate to at most one order.

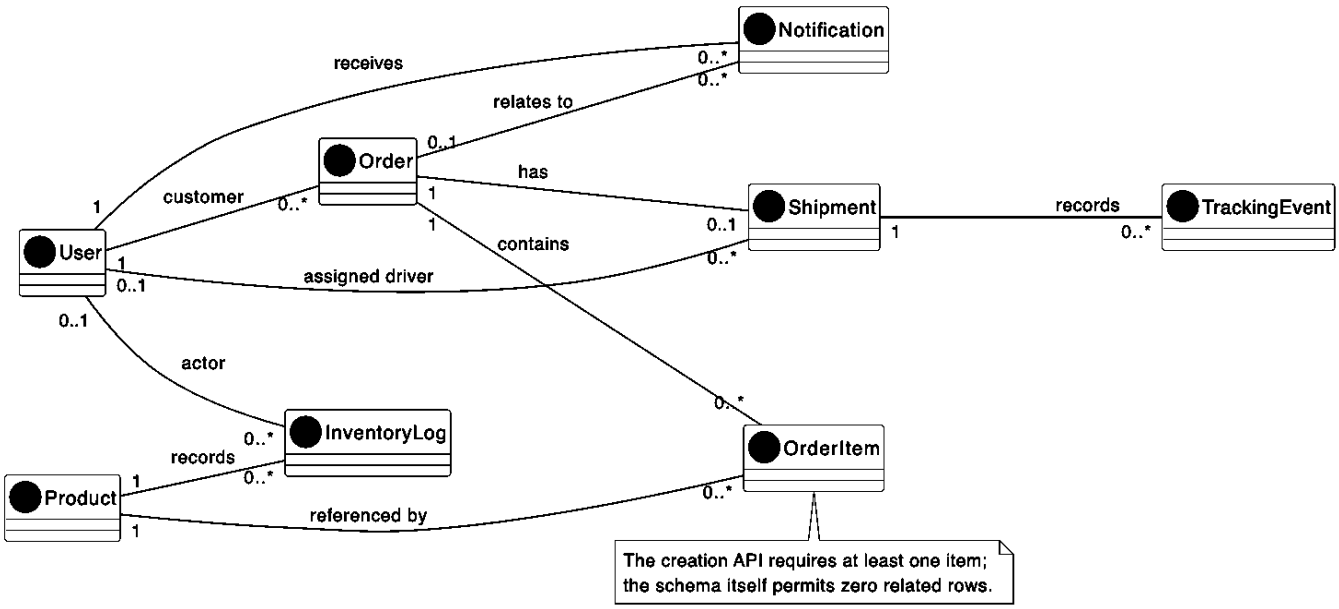


Figure 3. Data relationships. Source.

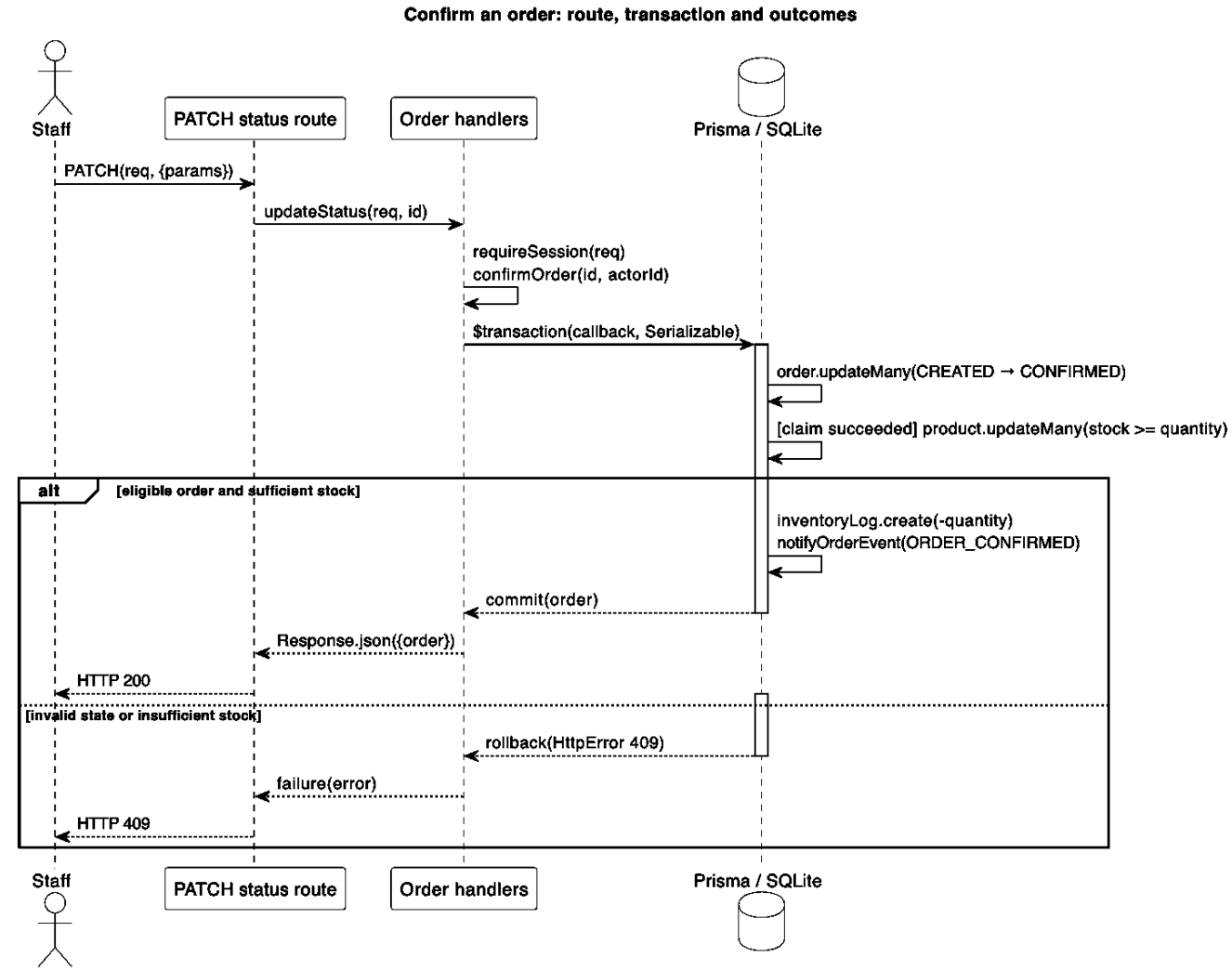


Figure 4. Confirmation transaction. Source.

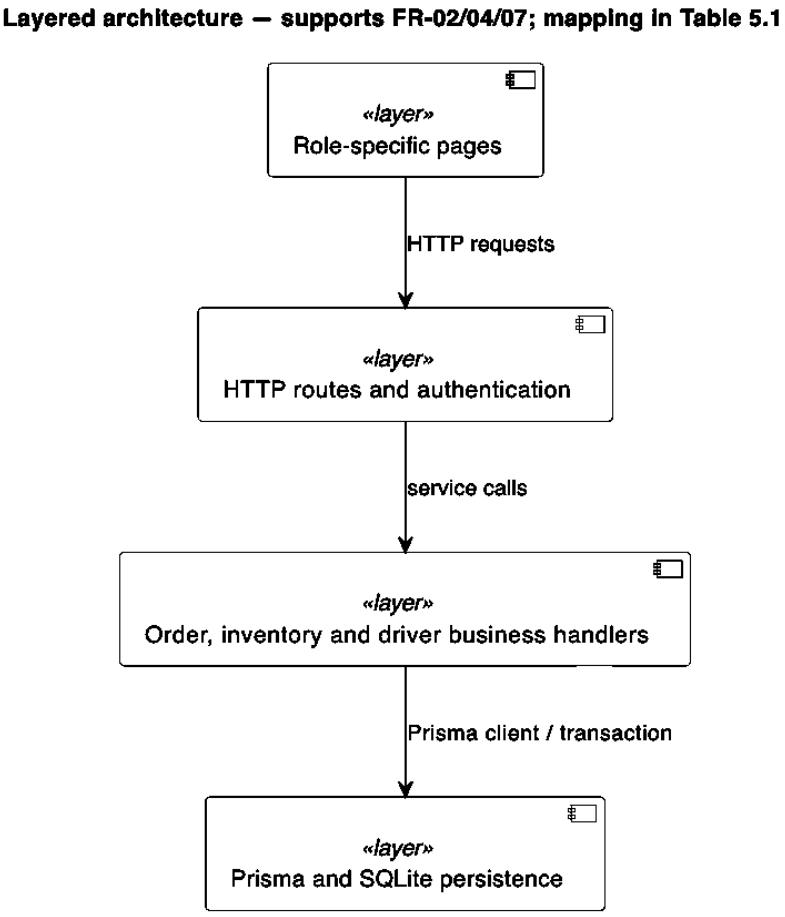


Figure 5. Four layers. Source.

5. Implementation

The confirmation transaction (order-api.ts:245–293) checks and reserves stock. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED; CANCELLED is reachable from CREATED or CONFIRMED, per the matrix below. Cancellations restock unless the order was CREATED, where stock was never reserved. Staff cancellation is restricted to CONFIRMED; a CREATED order is still the customer's to cancel.

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CONFIRMED	cancel, restock
Customer, staff	PICKING onwards	rejected

Design	Implementation	Verification
Guards: CREATED and sufficient stock	order-api.ts:250–253, 276–280	Orders
Confirmation transaction: writes state, stock, ledger and notifications atomically; rolls back on insufficient stock; repeats rejected (409)	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver
FR-03 numbering and tracking	create-order.ts:91, shipment-api.ts:62	Logistics chain
FR-04 ledger	order-api.ts:281, 398	Inventory
FR-05/06/07 handlers	notifications, reports and role gates follow the same handler pattern	Notifications, Reports, Auth

6. Verification

Verification checks requirements; validation exercises workflows (§8). CI records 38 unit and 140 integration checks; 2 integrity checks — the pipeline refuses to run without an explicit confirmation flag, and the packaged bundle digest must match the verified source (integrity checks); the integration log shows "tests 140 / pass 140", the regression log "Tests 38 passed (38)" and the walkthrough record "Browser acceptance passed: 26 checks".

In a local-loopback measurement (11 Nov; 30 requests per endpoint), API p95 3.2–13.6 ms and first-contentful-paint p95 64–96 ms against local-deployment targets of 2 s and 3 s.

7. Evolution

The 17 Oct 2026 review found a confirmation endpoint accepting repeat submissions and a missing ledger; the transaction and retest followed. The 10–11 Nov review corrected diagrams and traceability and replaced daily availability with CI-run checks (§3, §9).

8. Validation

The automated walkthrough exercises eleven core operations — create order; details; confirm; pick; ship; assign; pickup; delivery; report filter; export; selected-date export — each by its owning role, at most three in-app interface interactions each. All 26 checks passed (walkthrough record).

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagram bases diverged from the schema	Models corrected	10–11 Nov review	Diagra
Stock effects were recorded under FR-04	Moved stock effects to FR-02 (previously listed under FR-04); traceability uses the current numbering	17 Oct review	Traceal
Early register rows had thin triggers and outcomes	Register rewritten on 10–11 Nov	10–11 Nov review	Registe
Export took four scripted interactions	Added a direct selected-date export path (three in-app interface interactions); issue closed	11 Nov automated walkthrough	Issue #
The CI health record lacked a regression test	Reached 38 tests	11 Nov regression work	Test his
NFR-02 originally required daily availability from 08:00 through 24:00 local time	Replaced by CI-run checks (§3); local-deployment budgets recorded (§6)	10–11 Nov review	Require

10. Evidence

Traceability · Baseline record — the 12 Nov regression suite: 38 unit, 140 integration and 2 integrity checks; browser checks recorded 11 Nov.